

Dr. UPMA SINGH

Assistant Professor | Department of Electronics & Communication Engineering

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Maharaja Surajmal Institute of Technology (MSIT), Janakpuri, Delhi

Total Teaching Experience: 13+ Years (Serving since Jan 14, 2013)



EDUCATIONAL QUALIFICATIONS

- **Ph.D. (Electrical Engineering)** — Delhi Technological University (DTU), Delhi (2023)
Thesis: Modelling and Optimization of Hybrid Renewable Energy Systems and Applications
- **M.Tech. (Instrumentation & Control)** — Deenbandhu Chhotu Ram University of Science & Technology, Haryana (2012)
Dissertation: Intelligent approach for the prediction of wind energy
- **B.Tech. (Electronics & Communication)** — Vivekanand Institute of Science & Technology (UPTU) (2010)
- **Senior Secondary (12th Class)** — CBSE Board (2006)
- **Secondary School (10th Class)** — CBSE Board (2003)

EMPLOYMENT DETAILS

Employer / Organization	Designation	Period	Nature
Maharaja Surajmal Institute of Technology, Delhi	Assistant Professor	Jan 04, 2014 – Present	Permanent
Maharaja Surajmal Institute of Technology, Delhi	Assistant Professor	Aug 01, 2013 – Dec 14, 2013	Ad-hoc
Maharaja Surajmal Institute of Technology, Delhi	Assistant Professor	Jan 14, 2013 – May 25, 2013	Ad-hoc

AWARDS & HONORS

- Research Excellence Award (for Year 2025) – Maharaja Surajmal Institute of Technology (Awarded Feb 14, 2026)
- Research Excellence Award (for Year 2024) – Maharaja Surajmal Institute of Technology (Awarded Feb 14, 2026)
- Research Excellence Award (for Year 2022) – Maharaja Surajmal Institute of Technology
- Commendable Research Award – Delhi Technological University (DTU), Delhi

PATENTS

- **Title: Differential Equation Solver Computer Device** Application No: 6459678 | Field: Mathematics
- **Title: Wireless Electric Road for Charging Future EV Vehicles** Application No: 202321001634 | Field: Electrical Engineering

RESEARCH PUBLICATIONS (SELECTED INTERNATIONAL JOURNALS)

- **Upma Singh**, Saket Gupta, Mohd Zuhaib, and Marwan Ahmad Abdullah Sufyan. "Parameter-free hybrid RAO algorithm for reactive power optimization." *Scientific Reports* (2026). DOI: 10.1038/s41598-026-58206-6 (SCIE) IF: 4.9
- **Upma Singh**, Saket Gupta, Hasmat Malik, and Vinay Kumar Jadoun. "A comprehensive critical analysis of the African vulture optimization algorithm and its applications, challenges and future directions." *Discover Applied Sciences* (2026). DOI: 10.1007/s42452-026-08669-z (ECSE) IF: 3.8
- Saket Gupta, Neeraj Kumar, Mohd Zuhaib, Marwan Ahmad Abdullah Sufyan, **Upma Singh**, Sandeep Sharma, Vijay Bhuria, and Ankita Gupta. "Parameter-free optimization algorithm effective and precise solution of the optimal power flow problem." *Scientific Reports* (2026). DOI: 10.1038/s41598-026-48197-9 (SCIE) IF: 4.9
- **Upma Singh**, Nisha Singh, Hasmat Malik, and Mohammad Asef Hossaini. "Photovoltaic Power Forecasting: A Review on Models and Future Research Directions." *IET Renewable Power Generation* 20, no. 1 (2026): e70186. DOI: 10.1049/rpg2.70186 (SCIE) IF: 2.9
- Sandeep Singh, **Upma Singh**, Nisha Singh, Anjali Baliyan, Sachin Dhull, Himani Tomar, Neetu, "Advanced Predictive Analytics for Used Car Pricing Through Convolutional Neural Networks", *International Journal of Environmental Sciences*, Vol. 11 No. 22s, 2025. ISSN: 2229-7359, DOI: 10.64252/9k893k88 (SCOPUS)
- **Upma Singh**, Shekhar Singh, Saket Gupta, Majed A. Alotaibi, and Hasmat Malik. "Forecasting rooftop photovoltaic solar power using machine learning techniques." *Energy Reports* 13 (2025): 3616-3630. DOI: 10.1016/j.egyr.2025.03.005 (SCIE) IF: 5.1
- Shekhar Singh and **Upma Singh**. "Improving solar power forecast from meteorological parameters with regression models." *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* 46, no. 1 (2024): 2405-2420. DOI: 10.1080/15567036.2024.2307390 (SCIE) IF: 2.2
- **Upma Singh**, Sandeep Singh, Tripti Rathee, and Manav Vaish, "Customer Retention Modeling Over the OTT Platform using Machine Learning," *Indian Journal of Science & Technology*, 17(42), pp.4365-4371, October, 2024. DOI: 10.17485/IJST/v17i42.2619 (UGC-CARE)
- **Upma Singh**, Rachit Joon, Nikhil, Akhil Sherawat, "Inventory Sales and Demand Forecasting Using Machine Learning", *International Journal of Scientific Research in Engineering and Management (IJSREM)*, Volume: 08 Issue: 12 | Dec – 2024, SJIF Rating: 8.448, ISSN: 2582-3930. DOI: 10.55041/IJSREM40308 (UGC-CARE) IF: 8.1
- **Upma Singh** and M. Rizwan, "Analysis of wind turbine dataset and machine learning based forecasting in SCADA-system," *Journal of Ambient Intelligence and Humanized Computing*, 2022, pp.1-10. DOI: 10.1007/s12652-022-03878-x (SCOPUS) IF: 5.09
- **Upma Singh** and M. Rizwan, "Enhancing wind power forecasting from meteorological parameters using machine learning models," *Journal of Renewable and Sustainable Energy*, vol. 14, 063302, 2022, pp.1-10. DOI: 10.1063/5.0117662 (SCIE) IF: 2.3
- **Upma Singh**, Mohammad Rizwan, Hasmat Malik, and Fausto Pedro García Márquez, "Wind energy scenario, success and initiatives towards renewable energy in India—A review," *Energies* 15, no. 6 (2022): 2291. DOI: 10.3390/en15062291. (SCIE) IF: 3.2
- **Upma Singh**, Mohammad Rizwan, Muhannad Alaraj, and Ibrahim Alsaidan, "A machine learning-based gradient boosting regression approach for wind power production forecasting: a step towards smart grid environments," *Energies* 14, no. 16 (2021): 5196. DOI: 10.3390/en14165196 (SCIE) IF: 3.2
- **Upma Singh** and Mohammad Rizwan, "SCADA system dataset exploration and machine learning based forecast for wind turbines," *Results in Engineering*, 2022, p.100640. DOI: 10.1016/j.rineng.2022.100640 (ESCI) IF: 7.9
- **Upma Singh** and Mohammad Rizwan, "A Systematic Review on Selected Applications and Approaches of Wind Energy Forecasting and Integration," *Springer, Journal of The Institution of Engineers (India): Series B*, vol. 102, no. 5 (2021): 1061-1078. DOI: 10.1007/s40031-021-00618-1 (SCOPUS) IF: 2.31

- **Upma Singh and M. Rizwan**, "A feasibility study and cost-benefit analysis of an off-grid hybrid system for a rural area electrification," Solar Compass, vol. 3 (2022): 100031. DOI: 10.1016/j.solcom.2022.100031 (SCOPUS) IF: 5.63

PAPERS PUBLISHED IN CONFERENCE PROCEEDINGS

- **Upma Singh**, Nisha Singh, Anjali Baliyan, Vinshu Rathee, "A Comparative Performance Evaluation of Regression Models for Photovoltaic Power Forecasting," National Conference on Integrating Technologies, Ideas and Disciplines for Engineering Innovation, 2025.
- Anjali Baliyan, **Upma Singh**, Jasmine Chhikara, Shilpi, Yash, Karanjeet, "Evolution of MIMO Antennas: From 2G to 6G Wireless Systems," National Conference on Integrating Technologies, Ideas and Disciplines for Engineering Innovation, 2025.
- Sonia, **Upma Singh**, Aditya, Janvi, Jyoti, "Voice Biometric Attendance System using Speaker Recognition," National Conference on Integrating Technologies, Ideas and Disciplines for Engineering Innovation, 2025.
- **Upma Singh** and M. Rizwan, "Error Evaluation of Short-Term Wind Power Forecasting Models," 3rd International Conference on Inventive Computation and Information Technologies (ICICIT-2021) 12-13 August 2021, pp. 541-559. Springer, Singapore, 2022. DOI: 10.1007/978-981-16-6723-7_41
- **Upma Singh** and M. Rizwan, "Comparative Study of Machine Learning Techniques to Forecast Short-Term Wind Power," 2nd International Conference on Smart Data Intelligence (ICSMDI 2022), Trichy, Tamil Nadu, India, 11-12th April 2022, pp. 1-11. Springer, Book: Smart Data Intelligence. DOI: 10.1007/978-981-19-3311-0
- **Upma Singh** and M. Rizwan, "Analysis of Fuzzy Logic, ANN and ANFIS based Models for the Forecasting of Wind Power," Proceedings of 2018 2nd IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES), October 22-24, 2018, IEEE Xplore, pp. 1-7, IEEE, 2018. DOI: 10.1109/ICPEICES.2018.8897445
- M. Rizwan, Sanju Saini, and **Upma Singh**, "Prediction of Wind Energy using Intelligent Approach", IEEE India International Conference on Power Electronics (IICPE 2012), D.T.U, New Delhi.
- Narendra Kumar, S. K. Agarwal, **Upma Singh** and Vedpal, "Control of Static VAR System using Auxiliary Signal with IMDU for Improving the Transient Performance of Series Compensated Transmission Lines", IEEE National Electrical Engineering Conference, New Delhi, Dec. 2011.
- Narendra Kumar, Narender Hooda, Darshna Hooda, Manisha and **Upma Singh**, "Optimal Amount and Location of Series and Shunt Compensation for Enhancing Voltage Stability", IEEE National Electrical Engineering Conference, New Delhi, Dec. 2011.

ACADEMIC ENVIRONMENT & TEACHING

- **Courses Taught (UG Level):** Machine Learning, Analog Communications, Control Systems, Optoelectronics & Optical Communications, Electronic Devices, Digital Communication-II, Mobile Communication, Information Theory & Coding.
- **Laboratory Incharge:** Analog Communication Lab, Machine Learning Lab, Optical Communication Lab, Control System Lab at MSIT Delhi.
- **Academic & Administrative Roles:** Member of Library Committee, NDLI Club, College Magazine Committee, Time-Table Committee, and active participant in NBA accreditation process.

(Dr. UPMA SINGH)
Assistant Professor, ECE Dept.